

Boundary Value Testing

Boundary Value Testing (BVT) is a software testing technique where test cases are designed to test the "boundaries" or limits of input values. The idea is that errors often occur at the edges (boundaries) of input domains rather than in the middle.

Key Principles:

- **Boundary values** are the extreme values of input (maximum and minimum).
- Testing **just below** and **just above** the boundary values can uncover errors that are missed by normal testing.

Why Boundary Value Testing is Important:

- Software errors are often found at the boundary of input values.
- It's easier to test these edge cases because they are where most bugs appear.

Types of Boundary Conditions:

1. **Lower Boundary:** The minimum value the input can accept.
2. **Upper Boundary:** The maximum value the input can accept.
3. **Just Below the Boundary:** A value slightly lower than the minimum (or maximum).
4. **Just Above the Boundary:** A value slightly higher than the minimum (or maximum).

Steps for Boundary Value Testing:

1. **Identify the valid range** for the input (e.g., age: 18-60).
2. **Determine the boundary values** (e.g., 18 and 60).
3. **Create test cases** using:
 - The boundary values themselves (e.g., 18 and 60).
 - Values just below the boundary (e.g., 17 and 61).
 - Values just above the boundary (e.g., 19 and 59).

Example 1: Age Validation (Valid Range: 18 to 60)

- **Valid Range:** 18 to 60 years.
- **Boundary Values:**
 - Lower Boundary: 18
 - Upper Boundary: 60

Test Cases:

- **Test Case 1:** Input = **18** (Valid input)
 - Expected Output: **Valid** (age is within the acceptable range)
- **Test Case 2:** Input = **60** (Valid input)
 - Expected Output: **Valid** (age is within the acceptable range)
- **Test Case 3:** Input = **17** (Invalid input)
 - Expected Output: **Error message** (age must be 18 or above)
- **Test Case 4:** Input = **61** (Invalid input)
 - Expected Output: **Error message** (age must be 60 or below)
- **Test Case 5:** Input = **19** (Valid input)
 - Expected Output: **Valid** (age is within the acceptable range)
- **Test Case 6:** Input = **59** (Valid input)
 - Expected Output: **Valid** (age is within the acceptable range)

Example 2: PIN Code Validation (Valid Range: 4 digits)

- **Valid Range:** 4-digit PIN code (from 1000 to 9999).
- **Boundary Values:**
 - Lower Boundary: 1000
 - Upper Boundary: 9999

Test Cases:

- **Test Case 1:** Input = **1000** (Valid input)
 - Expected Output: **Valid PIN**
 - **Test Case 2:** Input = **9999** (Valid input)
 - Expected Output: **Valid PIN**
 - **Test Case 3:** Input = **999** (Invalid input, below the boundary)
 - Expected Output: **Error message** (PIN must be a 4-digit number)
 - **Test Case 4:** Input = **10001** (Invalid input, above the boundary)
 - Expected Output: **Error message** (PIN must be a 4-digit number)
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Example 3: Temperature Range for a Device (Valid Range: -10°C to 50°C)

- **Valid Range:** Temperature input between -10°C to 50°C.
- **Boundary Values:**
 - Lower Boundary: -10°C
 - Upper Boundary: 50°C

Test Cases:

- **Test Case 1:** Input = **-10°C** (Valid input)
 - Expected Output: **Device works**
- **Test Case 2:** Input = **50°C** (Valid input)
 - Expected Output: **Device works**
- **Test Case 3:** Input = **-11°C** (Invalid input, below the boundary)
 - Expected Output: **Error message** (temperature below -10°C)
- **Test Case 4:** Input = **51°C** (Invalid input, above the boundary)
 - Expected Output: **Error message** (temperature above 50°C)

Scenario-Based Exam Questions on Boundary Value Testing:

1. Scenario:

- A web application accepts user age input with the following valid range: 21 to 65 years.
- Write the boundary test cases for the age input.

Answer:

- **Boundary values:** 21 and 65.
- **Test cases:**
 - Test Case 1: Age = 21 (Valid)
 - Test Case 2: Age = 65 (Valid)
 - Test Case 3: Age = 20 (Invalid, below the boundary)
 - Test Case 4: Age = 66 (Invalid, above the boundary)

2. Scenario:

- A ticket booking system accepts bookings for events with seat numbers from 1 to 100.
- Identify and list test cases for Boundary Value Testing.

Answer:

- **Boundary values:** 1 (lower boundary), 100 (upper boundary).
- **Test cases:**
 - Test Case 1: Seat number = 1 (Valid)
 - Test Case 2: Seat number = 100 (Valid)
 - Test Case 3: Seat number = 0 (Invalid, below the boundary)
 - Test Case 4: Seat number = 101 (Invalid, above the boundary)

3. Scenario:

- A system accepts user ratings between 1 and 5 (inclusive).
- List the boundary test cases and explain the expected results.

Answer:

- **Boundary values:** 1 and 5.
- **Test cases:**
 - Test Case 1: Rating = 1 (Valid)
 - Test Case 2: Rating = 5 (Valid)
 - Test Case 3: Rating = 0 (Invalid, below the boundary)
 - Test Case 4: Rating = 6 (Invalid, above the boundary)

4. Scenario:

- You are testing an online form where the password must be between 8 and 12 characters.
- List the boundary test cases for the password length validation.

Answer:

- **Boundary values:** 8 (minimum) and 12 (maximum).
- **Test cases:**
 - Test Case 1: Password = "abcdefgh" (Valid, 8 characters)
 - Test Case 2: Password = "abcdefghijkl" (Valid, 12 characters)
 - Test Case 3: Password = "abcdefg" (Invalid, below 8 characters)
 - Test Case 4: Password = "abcdefghijklm" (Invalid, above 12 characters)

Boundary Value Testing focuses on testing the extreme values (boundaries) of an input range. By testing at the boundary, just below it, and just above it, we can uncover errors that might occur at these critical points. This is a highly effective and simple testing technique that can significantly improve software reliability.